MIKE3 Hydrodynamic model of Øresund

This repository contains practical and background information for the WaterBench case.

The dataset can be cited as:

> DHI. (2026). Hydrodynamic 3D model of the Øresund: MIKE3 model setup, outputs and observation data [Data set]. Zenodo. https://doi.org/10.5281/zenodo.20284684

See the LICENSE file for usage conditions.

Disclaimer

> This model setup and dataset are provided for educational, benchmarking, and research purposes only.
> The model configuration is simplified and may not be fully calibrated or validated for operational use.
> Results must not be used for real-time forecasting, infrastructure design, safety-critical decisions, or regulatory purposes.
> The dataset does not represent best-practice modelling standards at DHI and should be treated as a controlled benchmark case.

Intended Use

The dataset is aimed at supporting educational, research, and exploratory activities, such as:

- Development and benchmarking of hydrodynamic or hydrological models
- Testing model calibration and validation workflows
- Machine learning and surrogate modelling
- Extreme event forecasting studies
- Uncertainty quantification experiments

Folder Structure

```
- README.md
- LICENSE
- model/
    + model setup files
    + simulation logs
- input/
    + boundary conditions
    + meteorological forcings
```

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    + dispersion files
    + initial conditions
    + mesh file
- observations/
    + in-situ observations
    + satellite data
- code/
    + notebooks for data exploration
    + validation scripts
- figures/
- output_sample/
    + reduced model outputs for testing
- output/
    + full simulation results (only on Zenodo)
```

Background

This folder contains data for a simulation with the hydrodynamic MIKE 3 Flow Model FM in the Øresund Strait between Denmark and Sweden. It is an extension of a 2D setup published in 2024 (https://doi.org/10.5281/zenodo.14160710). The repository contains the setup file, input data and observation data, as well as simulation result data. The simulation result data is available in Zenodo. The data supplied in this repository covers the period from 2023-01-01 to 2025-12-31.

MIKE 3 Flow Model FM

MIKE 3 Flow Model FM is a 3D modelling system using a flexible mesh to discretize the horizontal dimension and a combination of a sigma and a Cartesian coordinate system for the vertical direction. The modelling system has been developed for applications within oceanographic, coastal and estuarine environments. The governing equations are the incompressible Navier-Stokes equations subject to the assumptions of Boussinesq (density variations are small enough to be ignored, except in gravity term) and with the free surface described by a height function. The current model setup also includes the transport equations for temperature and salinity, which is useful e.g. for ecological modelling or modelling of sediment transport. More information can be found in the documentation.

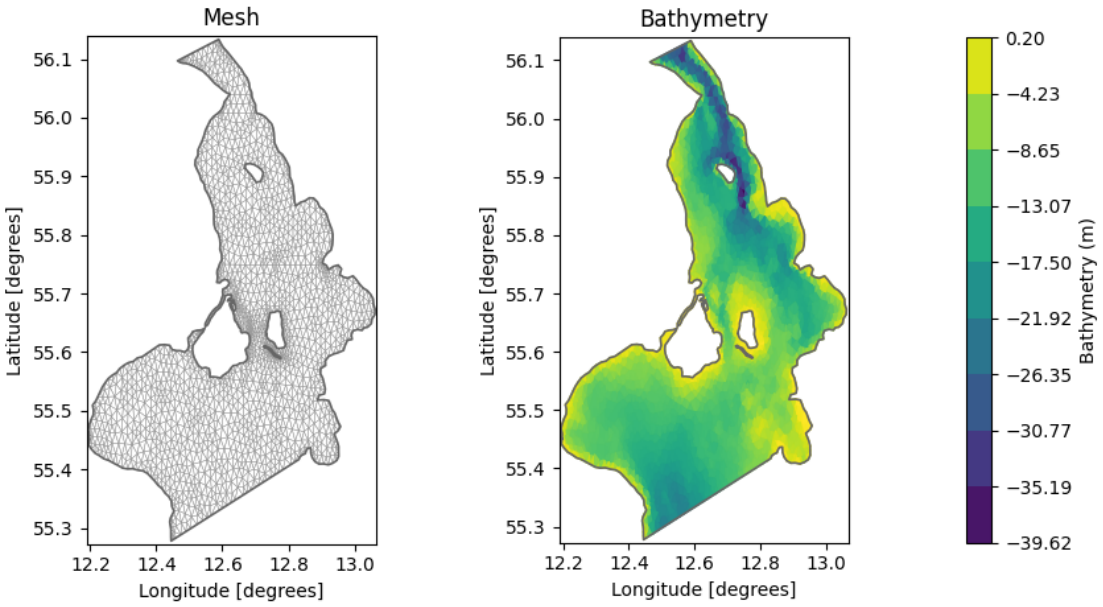
Since all data, including output from a simulation, is given, it is not necessary to run the MIKE simulation. However, it is possible, since a setup file is supplied as well as all of the input data. Running the MIKE software requires a license. Read more about obtaining a license and about academic licensing. If you already have a license, read the step-by-step guide to learn how to run simulations.

3D model of Øresund

The coastal hydrodynamics of Øresund are primarily driven by the surge component introduced at the two open boundaries (one in North and one in South) with tidal ranges of 10-20 cm. The meteorological forcings such as air pressure and wind have only a minor effect. Modelling the hydrodynamics of this area is interesting for many reasons, e.g. surveillance of bacteria in bathing water and sediments around constructions and infrastructure.

Model Setup

The Øresund mesh consists of triangular elements in a higher resolution close to the coast than on deeper waters, and for each node a depth value is assigned by interpolating the EMODnet bathymetry data. The mesh is shown in the figure below. It consists of a total of 3320 elements covering the area of approximately 2000 km^2.



Forcings

The model is forced by meteorological inputs and boundary conditions supplied from a larger regional model. The meteorological inputs are:

- wind velocities and air pressure,
- air temperature and relative humidity,
- cloud coverage, precipitation and sea-ice cover.

The boundary conditions supply:

- vertical cross section of water currents, temperature and salinity on the North and South boundary,
- water levels on the north and south boundary.

The outputs from the model are given as:

- a 3D timeseries of the current velocities (Eastward, Northward, and vertical directions), density, temperature and salinity.
- a 2D timeseries of the water surface elevations and depth-averaged current velocities.

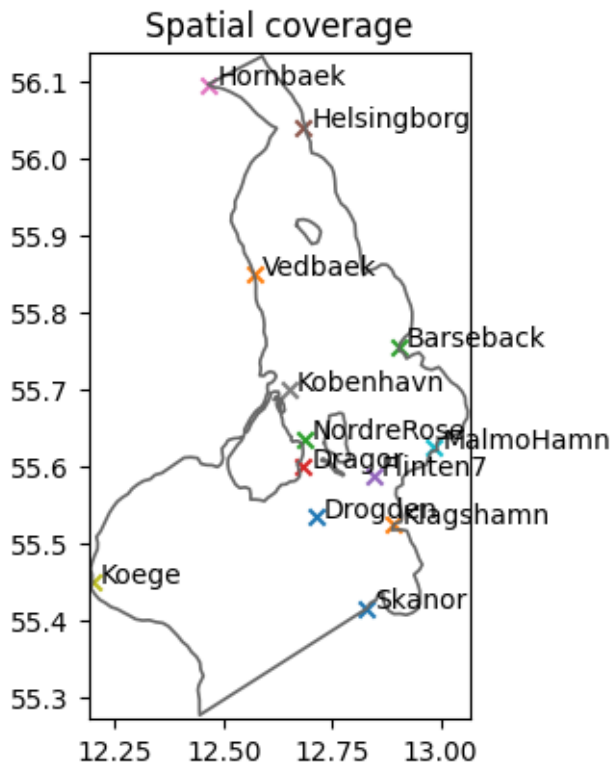
The 2D timeseries contains values for each mesh element, whereas the 3D timeseries also covers the coordinates in the vertical direction.

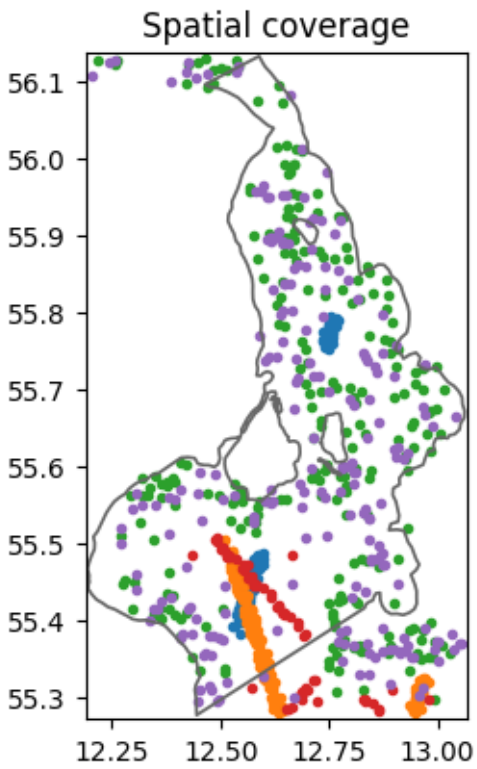
Validation of model outputs

The outputs of the model are validated against in-situ measurement as well as altimetry data from satellites. The validation scripts can be found in the ./code/ folder.

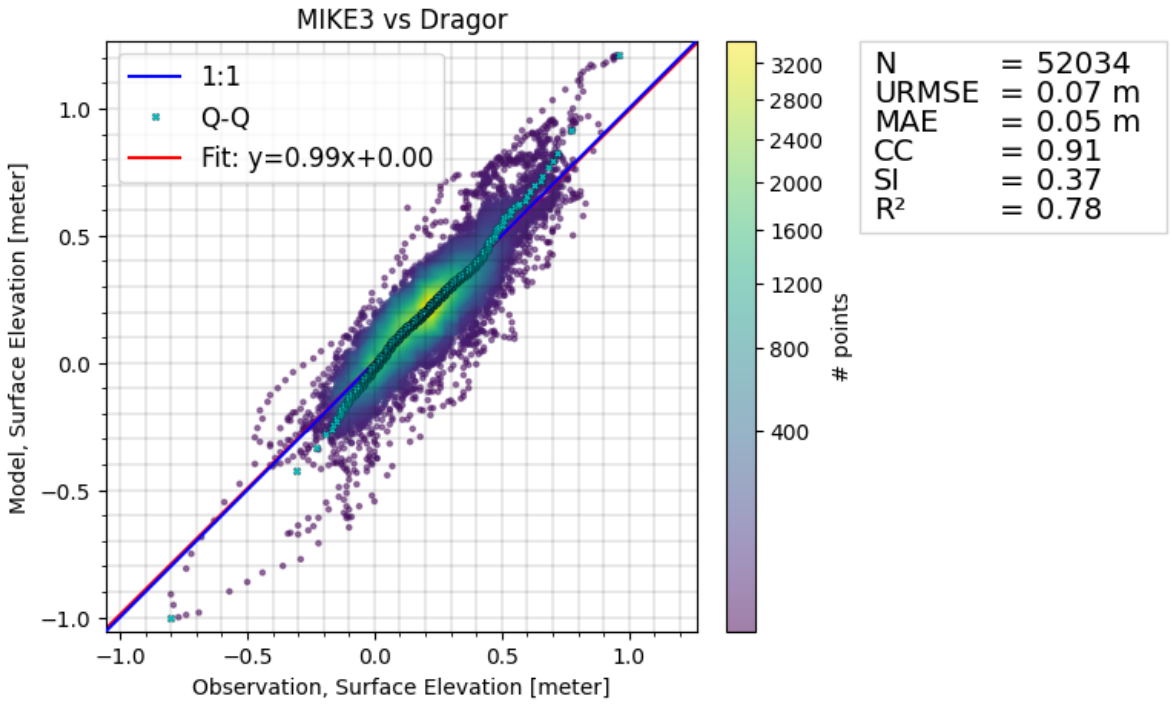
Validation of 2D outputs

The leftmost figure below shows the location of the in-situ locations, and the rightmost figure shows the coverage of the altimetry data.

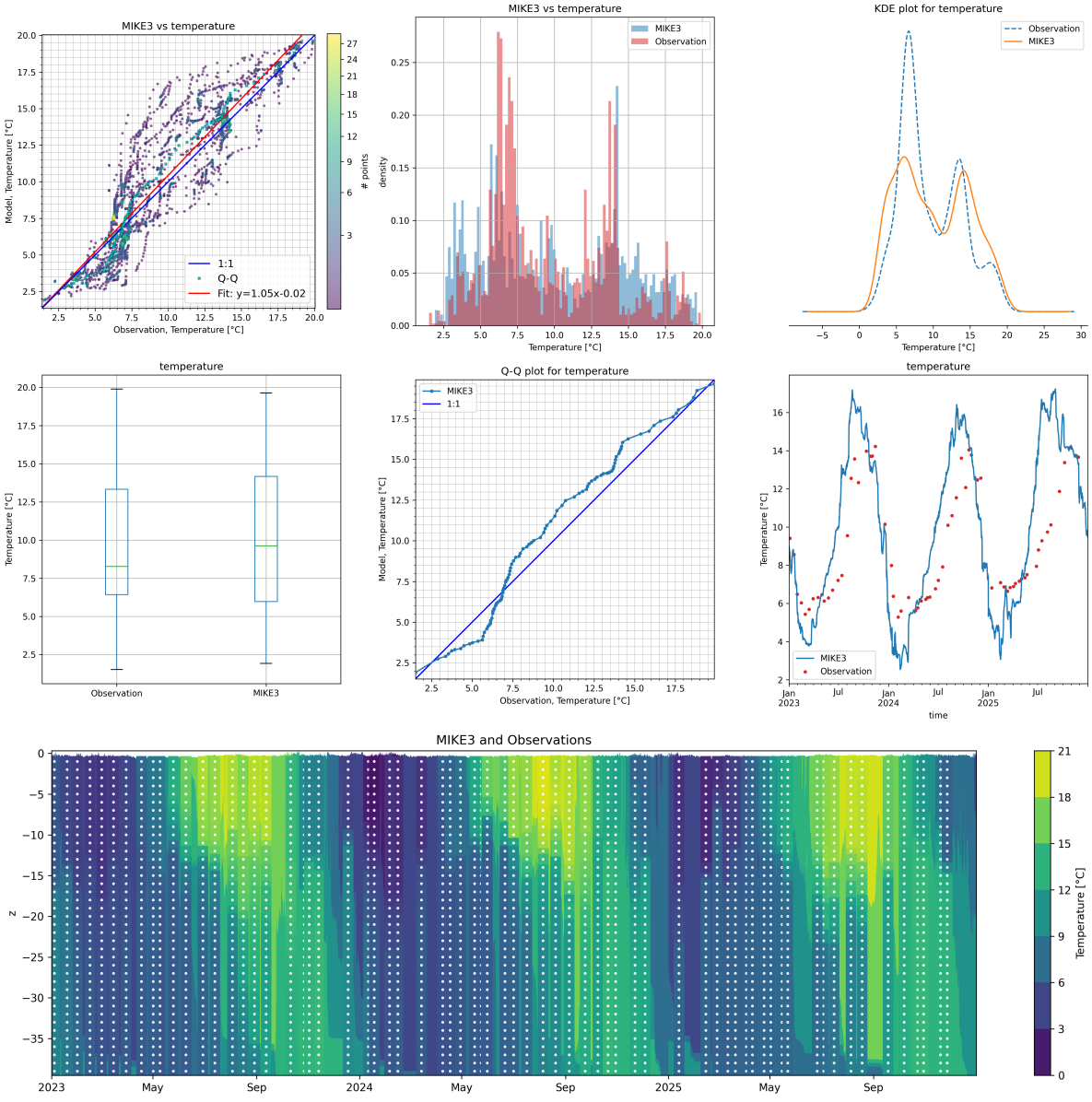




Below, the comparison between the observational data from one such measurement station and MIKE 21 is shown. The production of these and many more plots is found in the notebooks code/model_validation_2D.ipynb.


MIKE3 vs Dragor
1:1
Q-Q
Fit: y=0.99x+0.00
Model, Surface Elevation [meter]
Observation, Surface Elevation [meter]
points
N = 52034
URMSE = 0.07 m
MAE = 0.05 m
CC = 0.91
SI = 0.37
R² = 0.78

Validation of 3D outputs



Data Sources

The following changes have been made to the data:

- Sea levels in measurement stations: Data has been filtered to only include "good" or "probably good" data points according to quality flags. The coordinates of some stations have been moved into the simulation domain, since they are sometimes located so close to land (e.g. in a harbor) that the mesh is not covering it.
- Current velocities in measurement stations: Data has been filtered to only include "good" or "probably good" data points according to quality flags. Some faulty observations have been removed manually (e.g. zero-values and outliers). The u and v components of the current velocities are calculated from the speed and directions. The data is given at different depths, so depth-averaged values have been calculated.
- Altimetry data: Data has been filtered to only include "good" data points according to quality flags.

Data	Source	Citation	License
Mesh	DHI.	Repository citation, see the top	CC BY 4.0
Bathymetry	EMODnet Bathymetry Digital Terrain Model (DTM) 2020.	EMODnet Bathymetry Consortium (2020). EMODnet Digital Bathymetry (DTM 2020).EMODnet Bathymetry Consortium https://doi.org/10.12770/bb6a87dd-e579-4036-abe1-e649cea9881a	CC BY 4.0
Dispersion coefficients	DHI	Repository citation, see the top	CC BY 4.0
Boundary condition	DHI's Danish Waters and Baltic Sea (DKBS) model.	DHI, "Water levels and currents on the boundaries of Øresund from the DKBS hydrodynamical model" (2024).	CC BY 4.0

Data	Source	Citation	License
Meteorological forcing	The ERA5 reanalysis of globate climate.	Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., Nicolas, J., Peubey, C., Radu, R., Rozum, I., Schepers, D., Simmons, A., Soci, C., Dee, D., Thépaut, J-N. (2023): ERA5 hourly data on pressure levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), DOI: 10.24381/cds.bd0915c6 (Accessed on 01-05-2026)	See full license here. Allowed to use and redistribute.
Sea levels in measurement stations	Copernicus Marine Service (CMEMS, or Copernicus Marine Environmental Monitoring Service). Downloaded from the Marine Data Store.	Copernicus Marine Service, "Baltic Sea- In Situ Near Real Time Observations" (2024), https://doi.org/10.48670/moi-00032.	See full license here. Allowed to use and redistribute for any purpose.
Current velocities in measurement stations	SMHI (Meteorological and Hydrological Institute of Sweden).	Meteorological and Hydrological Institute of Sweden, "Oceanographic observations" (2024), URL: https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-observationer-fran-havet/current (accessed: 2026-05-20)	Publicly available and free to use for all purposes. Read more
Temperature and salinity measurements	Danmarks Miljøportal	Danmarks Miljøportal (2026), URL: https://miljoeportal.dk/ (Accessed: 2026-05-20)	CC BY 4.0

Altimetry data

The altimetry data is extracted from a range of different sources. The data is based on satellite observations, and the data is downloaded using DHI's services for marine observation data. Data sources:

Satellite name	Owner	Link
Sentinel-3A (3a)	European Space Agency (ESA)	https://sentiwiki.copernicus.eu/web/s3-altimetry-instruments
Sentinel-3B (3b)	European Space Agency (ESA)	https://sentiwiki.copernicus.eu/web/s3-altimetry-instruments
Sentinel-6A (6a)	European commission, ESA, EUMETSAT, NASA and NOAA	https://sentinels.copernicus.eu/web/sentinel/missions/sentinel-6/data-products
CryoSat-2 (c2)	European Space Agency (ESA)	https://earth.esa.int/eogateway/catalog/cryosat-products
SARAL (sa)	French National Space Agency (CNES) and Indian Space Research Organisation (ISRO)	https://podaac.jpl.nasa.gov/dataset/ALTIKA_SARAL_L2_OST_XOGDR
Jason-3 (j3)	National Aeronautics and Space Administration (NASA) and French National Space Agency (CNES)	https://sealevel.jpl.nasa.gov/missions/jason-3/summary/